Aritra Mukherjee

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Course	College/University	Year	CGPA/%
Ph.D Scholar	IIT Kanpur	2026-	
M.Tech ECE	IIT Dhanbad	2024-2026	8.86/10
B.Tech ECE	Techno Main Salt Lake, MAKAUT	2018-2022	8.79/10

SCHOLASTIC ACHIEVEMENTS

- Ranked **3rd in M.Tech** (1st Yr) ECE at Optical communication & integrated photonics, IIT Dhanbad.
- Secured **AIR-2549** in Gate 2024 among 63,092 candidates.

WORK EXPERIENCE

- **Assistant System Engineer | Tata Consultancy Services (TCS), Kolkata** [Oct'22-Jun'24]
  - Worked on the development, testing, and maintenance of enterprise-level applications in an agile environment.
  - Collaborated with cross-functional teams to deliver scalable software solutions for client-specific requirements.
  - Involved in code development, debugging, and performance optimization using technologies such as Java and SQL.

TECHNICAL SKILLS

- **Programming Languages**: C, Python, Java
- **Tools & Technologies**: MATLAB, IBM Qiskit, Pandas, Numpy, Seaborn, Matplotlib

AREAS OF INTEREST

- **Optimization | Machine Learning | Deep Learning | Signal Processing for Communications**

RELEVANT COURSES

- **Optimization Theory & Techniques | Wireless Networks | Machine Learning | Linear Algebra | VLSI Signal Processing** [IIT-ISM]
- **Information Theory & Coding | Digital Comm. & Stochastic Processes | Digital Signal Processing | Analog Comm.**[MAKAUT]

PROJECTS

- **Fundamental Limits on Entanglement-Assisted Quantum Online Prediction | Dr. S. Mukhopadhyay & Dr. J. Kumar** [Nov'24-Ongoing]
  - Aiming to minimize the lower bound of risk/regret involved in the Quantum Online Learning setting.
  - Encompasses Quantum Games, Quantum Information Theory, and knowledge of Classical online Machine Learning.
- **Implementation of HHL Quantum Algorithm | MATLAB Lab Environment** [Academic]
  - Focused on solving systems of linear equations through quantum computing principles in photonic systems.
  - Simulating quantum gates and circuits using linear optics elements like beam splitters and phase shifters.
- **Enhancement of Yagi-Uda Antenna Using Microstrip Circuit** [Jan'21-Sep'21]
  - Achieved improved bandwidth and impedance matching by optimizing the microstrip feed and reflector-director configuration.
  - Utilized HFSS to model the antenna structure, perform electromagnetic simulations, and analyze parameters.

POSITIONS OF RESPONSIBILITY & CERTIFICATIONS

- **Class Representative | M.Tech ECE Batch, IIT (ISM) Dhanbad** [2024-2026]
  - Acting as a liaison between faculty and students via facilitating smooth communication and coordination.
  - Represented student concerns, organized academic discussions, and contributed to resolving administrative issues.
- **NPTEL Certifications**
  - Biomedical Signal Processing, Jan-Apr 2020 (12 week course), issued by IIT Kharagpur.
  - Introduction to Internet of Things, Jan-Apr 2020 (12 week course), issued by IIT Kharagpur.
  - Joy of Computing Using Python, Jan-Apr 2019 (12 week course), issued by IIT Madras.